Oefeningen "H6: Recurrente betrekkinge (uigewerkte oplossingen) Deel 2: Niet-homogene recurrente betre (wiskundige vragen)

```
-->     kill(all);
(%o0)   done
```

1) Oefening 10
a(n)-2a(n-1)-3a(n-2)=8n met a(1)=-6 en a(2)=4

STAP 1: a(n)-2a(n-1)-3a(n-2)=0 (homogeen)

```
-->     kv:r^2-2*r-3=0;
```

(kv) $r^2-2\,r-3=0$

```
-->     solve(kv);
```

(%o2) $[r=3,r=-1]$

```
-->     a_H(n):=alpha*(-1)^n+beta*3^n;
```

(%o3) $a_H(n):=\alpha\,(-1)^n+beta\,3^n$

STAP 2: particuliere oplossing
(voorstel: An+B, niet aanpassen want geen "overlap" met a_H)

```
-->     a_P(n):=A*n+B;
```

(%o4) $a_P(n):=A\,n+B$

```
-->     ingevuld:a_P(n)-2*a_P(n-1)-3*a_P(n-2)=8*n;
```

(ingevuld) $A\,n-2\,(A\,(n-1)+B)-3\,(A\,(n-2)+B)+B=8\,n$

```
-->     ingevuld:expand(%);
```

(ingevuld) $-4\,A\,n-4\,B+8\,A=8\,n$

Stelsel oplossen met wxMaxima

(opletten, laten inwerken na expand)

```
-->     vgl1:coeff(ingevuld,n,1);
```

(vgl1) $-4\,A=8$

```
-->     vgl2:coeff(ingevuld,n,0);
```

(vgl2) $8\,A-4\,B=0$

```
-->     solve([vgl1,vgl2]);
```

(%o9) $[[B=-4,A=-2]]$

```
-->     A:-2; B:-4;
(A)     -2
(B)     -4
```

```
-->     a_P(n);
(%o12)  -2 n-4
```

STAP 3: (voorlopige) eindoplossing

```
-->     a(n):=a_H(n)+a_P(n)$
        a(n);
```

(%o14) $beta\,3^n+\alpha\left(-1\right)^n-2\,n-4$

STAP 4: unieke eindoplossing met a(1)=-6 en a(2)=4

```
-->     vgl1:a(1)=-6;
```

(vgl1) $3\,beta-\alpha-6=-6$

```
-->     vgl2:a(2)=4;
```

(vgl2) $9\,beta+\alpha-8=4$

```
-->     solve([vgl1,vgl2]);
```

(%o17) $[[beta=1,\alpha=3]]$

```
-->     alpha:3; beta:1;
(alpha) 3
(beta)  1
```

```
-->     a(n);
```

(%o20) $3^n+3\left(-1\right)^n-2\,n-4$

```
-->     a(10);
(%o21)  59028
```

2) Oefening 11

a(n)-a(n-1)-2a(n-2)=4n met a(1)=-9 en a(2)=-1

```
-->     kill(all);
(%o0)   done
```

STAP 1: a(n)-a(n-1)-2a(n-2)=0 (homogeen)

```
-->     kv:r^2-r-2*1=0;
```

(kv) $r^2-r-2=0$

```
-->     solve(kv);
```

(%o2) $[r=2,r=-1]$

```
-->     a_H(n):=alpha*(-1)^n+beta*2^n;
```

(%o3) $a_H\left(n\right):=alpha\left(-1\right)^n+beta\,2^n$

STAP 2: particuliere oplossing
(voorstel: An+B)

```
-->      a_P(n):=A*n+B;
```
(%o4) $a_P(n):=A\,n+B$

```
-->      a_P(n)-a_P(n-1)-2*a_P(n-2)=4*n;
```
(%o5) $A\,n-A\left(n-1\right)-2\left(A\left(n-2\right)+B\right)=4\,n$

```
-->      ingevuld:expand(%);
```
(ingevuld) $-2\,A\,n-2\,B+5\,A=4\,n$

We kunnen het stelsel ook aflezen
(door te kijken naar de machten van n: eerste macht en nulde macht=constante

```
-->      stelsel:[-2*A=4,-2*B+5*A=0];
```
(stelsel) $[-2\,A=4,5\,A-2\,B=0]$

```
-->      solve(stelsel);
```
(%o8) $[[B=-5,A=-2]]$

```
-->      A:-2; B:-5;
```
(A) -2
(B) -5

```
-->      a_P(n);
```
(%o11) $-2\,n-5$

STAP 3: (voorlopige) eindoplossing

```
-->      a(n):=a_H(n)+a_P(n);
```
(%o12) $a\left(n\right):=a_H\left(n\right)+a_P\left(n\right)$

```
-->      a(n);
```
(%o13) $beta\,2^n+alpha\left(-1\right)^n-2\,n-5$

STAP 4: (definitieve) eindoplossing

```
-->      vgl1:a(1)=-9;
```
(vgl1) $2\,beta-alpha-7=-9$

```
-->      vgl2:a(2)=-1;
```
(vgl2) $4\,beta+alpha-9=-1$

```
-->      solve([vgl1,vgl2]);
```
(%o16) $[[beta=1,alpha=4]]$

```
-->      alpha:4; beta:1;
```
(alpha) 4
(beta) 1

```
-->      a(n);
```
(%o19) $2^n+4\left(-1\right)^n-2\,n-5$

```
-->     a(10);
(%o20)  1003
```

3) oefening 12

3.1 (a) differentievgl: a(n)-2a(n-1)+a(n-2)=6n-12n²

```
-->     kill(all);
(%o0)   done
```

STAP 1: a(n)-2a(n-1)+a(n-2)=0(homogeen)

```
-->     kv:r^2-2*r+1=0;
(kv)    r^2-2 r+1=0
```

$$r^2-2\,r+1=0$$

```
-->     solve(kv);
(%o2)   [r=1]
```

$$[r=1]$$

```
-->     a_H(n):=alpha*1^n+beta*1^n*n;
(%o3)   a_H(n):=alpha 1^n+beta 1^n n
```

$$a_H(n):=alpha\,1^n+beta\,1^n\,n$$

```
-->     a_H(n);
(%o4)   beta n+alpha
```

$$beta\,n+alpha$$

STAP 2: particuliere oplossin
(kopie van RL dus voorstel: (An²+Bn+C)n²
want vertrekken van An²+Bn+C maar *n en *n (om verschillend te maken van a_H

```
-->     a_P(n):=(A*n^2+B*n+C)*n^2;
(%o5)   a_P(n):=(A n^2+B n+C) n^2
```

$$a_P(n):=\left(A\,n^2+B\,n+C\right)n^2$$

```
-->     a_P(n)-2*a_P(n-1)+a_P(n-2)=6*n-12*n^2;
(%o6)   n^2 (A n^2+B n+C)-2 (B (n-1)+A (n-1)^2+C) (n-1)^2+
(B (n-2)+A (n-2)^2+C) (n-2)^2=6 n-12 n^2
```

$$n^2\left(A\,n^2+B\,n+C\right)-2\left(B\left(n-1\right)+A\left(n-1\right)^2+C\right)\left(n-1\right)^2+\left(B\left(n-2\right)+A\left(n-2\right)^2+C\right)\left(n-2\right)^2=6\,n-12\,n^2$$

```
-->     ingevuld:expand(%);
(ingevuld) 12 A n^2+6 B n-24 A n+2 C-6 B+14 A=6 n-12 n^2
```

$$12\,A\,n^2+6\,B\,n-24\,A\,n+2\,C-6\,B+14\,A=6\,n-12\,n^2$$

```
-->     vgl1:coeff(ingevuld,n,0);
(vgl1)  2 C-6 B+14 A=0
```

$$2\,C-6\,B+14\,A=0$$

```
-->     vgl2:coeff(ingevuld,n,1);
(vgl2)  6 B-24 A=6
```

$$6\,B-24\,A=6$$

```
-->     vgl3:coeff(ingevuld,n,2);
(vgl3)  12 A=-12
```

$$12\,A=-12$$

```
-->     solve([vgl1,vgl2,vgl3]);
(%o11)  [[C=-2,B=-3,A=-1]]
```

$$[[C=-2,B=-3,A=-1]]$$

```
-->     A:-1; B:-3; C:-2;
(A)     -1
(B)     -3
(C)     -2
```

```
-->     a_P(n);
```

(%o15) $n^2\left(-n^2-3\,n-2\right)$

STAP 3: eindoplossing

```
-->     a(n):=a_H(n)+a_P(n);
```

(%o16) $a\left(n\right):=a_H\left(n\right)+a_P\left(n\right)$

```
-->     expand(a(n));
```

(%o17) $-n^4-3\,n^3-2\,n^2+beta\,n+alpha$

3.2 (b)
differentievgl: a(n)-2a(n-1)+a(n-2)=12n²

```
-->     kill(all);
(%o0)   done
```

STAP 1: a(n)-2a(n-1)+a(n-2)=0(homogeen)

```
-->     kv:r^2-2*r+1=0;
(kv)    r^2-2 r+1=0
```

```
-->     solve(kv);
(%o2)   [r=1]
```

```
-->     a_H(n):=alpha*1^n+beta*1^n*n;
```

(%o3) $a_H\left(n\right):=alpha\,1^n+beta\,1^n\,n$

```
-->     a_H(n);
(%o4)   beta n+alpha
```

STAP 2: particuliere oplossin
(kopie van RL dus voorstel: (An²+Bn+C)n²
want vertrekken van An²+Bn+C maar *n en *n (om verschillend te maken van a_H

```
-->     a_P(n):=(A*n^2+B*n+C)*n^2;
```

(%o5) $a_P\left(n\right):=\left(A\,n^2+B\,n+C\right)n^2$

```
-->     a_P(n)-2*a_P(n-1)+a_P(n-2)=12*n^2;
```

(%o6) $n^2\left(A\,n^2+B\,n+C\right)-2\left(B\left(n-1\right)+A\left(n-1\right)^2+C\right)\left(n-1\right)^2+\left(B\left(n-2\right)+A\left(n-2\right)^2+C\right)\left(n-2\right)^2=12\,n^2$

```
-->     ingevuld:expand(%);
```

(ingevuld) $12\,A\,n^2+6\,B\,n-24\,A\,n+2\,C-6\,B+14\,A=12\,n^2$

```
-->     vgl1:coeff(ingevuld,n,0);
```
(vgl1) $2\,C-6\,B+14\,A=0$

```
-->     vgl2:coeff(ingevuld,n,1);
```
(vgl2) $6\,B-24\,A=0$

```
-->     vgl3:coeff(ingevuld,n,2);
```
(vgl3) $12\,A=12$

```
-->     solve([vgl1,vgl2,vgl3]);
```
(%o11) $[[C=5,B=4,A=1]]$

```
-->     A:1; B:4; C:5;
```
(A) 1

(B) 4

(C) 5

```
-->     a_P(n);
```
(%o15) $n^2\left(n^2+4\,n+5\right)$

STAP 3: eindoplossing

```
-->     a(n):=a_H(n)+a_P(n);
```
(%o16) $a\left(n\right):=a_H\left(n\right)+a_P\left(n\right)$

```
-->     expand(a(n));
```
(%o17) $n^4+4\,n^3+5\,n^2+beta\,n+alpha$